

Tile-Based Quality Reconstruction for Short-Exposure Deep-Sky Imaging

Mathematical Specification v3.3.9, Soft Local Blending, Support-Aware OLA,
and an M31 Example Run

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Abstract

Tile-Based Quality Reconstruction (TBQR) is a reconstruction methodology for short-exposure deep-sky datasets that replaces binary best-frame selection with a continuous spatio-temporal quality model. Version 3.3.9 refines the mathematical core established in v3.3.6 with six substantive changes: (1) explicit separation of additive background subtraction and multiplicative photometric scaling in the normalization step; (2) a continuous star-support blend factor that replaces the hard STAR/STRUCTURE tile switch; (3) neighborhood-aware metric normalization that pools z-scores across tile neighborhoods; (4) edge-aware spatial regularization of local quality scores on the tile-neighborhood graph; (5) support-aware overlap-add that uses a per-pixel support mask instead of per-tile median/MAD renormalization; and (6) mass-preserving quality-weighted cluster aggregation in final stacking. The paper consolidates the mathematical definitions used by the implementation, documents the relevant functions and constraints, discusses the delta from v3.3.6, and grounds the method in a concrete M31 run. The example run processes 645 usable frames on a 3926×2322 canvas with 209 adaptive tiles and yields a median output FWHM of 5.108 px from a median seeing estimate of 12.351 px, corresponding to a measured improvement of 58.640%.

Keywords: astronomical image reconstruction, deep-sky imaging, short exposure stacking, robust estimation, background modeling, photometric color calibration

1 Introduction

Short-exposure deep-sky imaging intentionally acquires large numbers of comparatively noisy frames in order to average atmospheric fluctuations, tracking errors, and sensor noise over time. The classical answer to this variability is a best-frame strategy: rank frames by a global sharpness criterion, keep only the top subset, and discard the rest. This is effective in high-SNR planetary imaging, but it is a poor fit for deep-sky objects because discarded frames also

discard photons, and signal-to-noise ratio scales only with the square root of retained exposure time.

TBQR replaces binary selection by a continuous weighting model. The central premise is that image quality is not a single scalar attached to the whole frame; it is the product of global atmospheric conditions and local structural fidelity. A frame may be globally noisy but still contribute usefully in tiles that remain sharp, and vice versa. The reconstruction target is therefore a per-pixel, per-tile weighted estimate assembled from all geometrically valid frames.

Version 3.3.9 builds on the v3.3.6 core and sharpens six areas. First, the normalization step now explicitly separates additive sky-background subtraction from multiplicative photometric scaling, preventing sky background from being the sole determinant of the photometric scale. Second, the hard STAR/STRUCTURE tile classification is replaced by a continuous star-support blend factor, so tiles with intermediate star counts receive a smooth interpolation between the two quality models rather than a discontinuous jump. Third, z-scores for local tile metrics now pool statistics across a spatial neighborhood of tiles, improving robustness for tiles with limited per-frame sample counts. Fourth, a graph-based spatial regularization step smooths local quality scores across the tile grid using edge-aware affinity weights, preventing neighboring tiles from diverging into incompatible weight regimes. Fifth, the overlap-add stage drops the per-tile median/MAD renormalization and instead uses a per-pixel support mask that excludes fill-zero contributions from coverage boundaries. Sixth, cluster aggregation in the final stacking step incorporates a cluster mass term so that clusters with more total atmospheric weight exert correspondingly more influence.

2 Invariants, Notation, and Function Catalog

2.1 Binding Invariants

The methodology is constrained by four invariants that have been unchanged since v3.3.6.

1. **No quality-based frame selection.** Frames are not discarded because they are globally worse than other frames.
2. **Geometric validity gating is allowed.** Frames may be rejected when registration is physically implausible, for example under reflection, catastrophic scale error, or extremely poor registration correlation.
3. **Strict linearity in the reconstruction core.** The core uses only linear or affine operations on

linear pixel values. Robust/statistical nonlinearities (MAD, clipping, sigma clipping) are allowed as auxiliary steps but do not alter the linearity semantics of pixel reconstruction.

4. **CFA-aware handling is preserved where relevant.** Registration and channel separation must not break Bayer phase semantics.

2.2 Notation

Indices and symbols follow the v3.3.9 specification:

- f : frame index; t : tile index; c : channel index; p : pixel
- $I_{f,c}^{raw}(p)$: raw linear input frame
- $B_{f,c}$: global additive background (pre-normalization)
- $P_{f,c}$: global photometric scale (new in v3.3.9)
- $I_{f,c}(p)$: normalized input frame
- $\sigma_{f,c}$: global noise (post-normalization)
- $E_{f,c}$: global gradient energy (post-normalization)
- $Q_{f,c}$: global quality index
- $G_{f,c}$: global weight; k_{global} : global weight scale
- η_t : tile star-support blend factor (new in v3.3.9)
- $\tilde{z}(\cdot)$: neighborhood-aware z-score (new in v3.3.9)
- $Q_{f,t,c}^{local}$: local tile quality index
- $L_{f,t,c}$: local tile weight; k_{local} : local weight scale (new)
- $W_{f,t,c}$: effective reconstruction weight
- $M_{t,c}(p)$: per-pixel support mask for OLA (new in v3.3.9)
- $M_{k,c}$: cluster mass (new in v3.3.9)
- $R_{t,c}(p)$: reconstructed tile

Notation note (v3.3.9): The symbols α, β, γ refer exclusively to the global quality index weights (Section 3.2). The BGE auto-tuning objective uses the distinct symbols α_f (flatness weight) and β_r (roughness weight) to avoid collision.

3 Core Reconstruction Mathematics

3.1 Global Linear Normalization

The normalization step is refactored in v3.3.9 to explicitly separate additive background removal from multiplicative photometric scaling. The sequence is:

$$B_{f,c} = \text{median}(I_{f,c}^{raw}), \quad (1)$$

$$J_{f,c}(p) = I_{f,c}^{raw}(p) - B_{f,c}, \quad (2)$$

$$P_{f,c} = \text{photometric_scale}(J_{f,c}), \quad (3)$$

$$I_{f,c}(p) = \frac{J_{f,c}(p)}{\max(P_{f,c}, \varepsilon_{scale})}. \quad (4)$$

The binding constraint is that $P_{f,c}$ must not be defined solely by the sky-background level $B_{f,c}$. The recommended photometric scale is ensemble stellar aperture flux relative to the reference frame. When stellar flux is unavailable but all frames share identical exposure times and throughput, the deterministic fallback $P_{f,c} = 1$ is admissible. Noise $\sigma_{f,c}$ and gradient energy $E_{f,c}$ are measured on the normalized data $I_{f,c}$.

This contrasts with v3.3.6, where the normalization was a single division $I_{f,c}(p) = I_{f,c}^{raw}(p) / \max(B_{f,c}, \varepsilon_{bg})$ and therefore conflated additive background level with multiplicative scale.

3.2 Global Quality Index and Adaptive Weights

The global quality index is unchanged from v3.3.6:

$$Q_{f,c} = \alpha(-z(B_{f,c})) + \beta(-z(\sigma_{f,c})) + \gamma z(E_{f,c}), \quad (5)$$

subject to $\alpha + \beta + \gamma = 1$, with default static weights $(\alpha, \beta, \gamma) = (0.4, 0.3, 0.3)$. The global weight is

$$G_{f,c} = \exp(k_{global} \text{clip}(Q_{f,c}, -3, +3)), \quad (6)$$

with default $k_{global} = 1.0$. Optional adaptive weighting is an extension whose utility criterion and tie-break rules must be documented.

3.3 Adaptive Tile Geometry

Tile size follows the seeing estimate F (FWHM in pixels):

$$T_0 = sF, \quad T = \lfloor \text{clip}(T_0, T_{min}, \lfloor \min(W, H)/D \rfloor) \rfloor, \quad (7)$$

$$O = \lfloor o_{clipped} \cdot T \rfloor, \quad S = T - O, \quad (8)$$

with $o_{clipped} = \text{clip}(o, 0, 0.5)$ and guards $F > 0$, $T_{min} \geq 16$, $S > 0$. A deterministic compact-tile fallback ($T = \min(W, H)$, $O = 0$) activates when $\lfloor \min(W, H)/D \rfloor < T_{min}$.

3.4 Local Tile Metrics and Effective Weights (v3.3.9)

3.4.1 Soft Star-Support Blend

The hard STAR/STRUCTURE switch present in v3.3.6 is replaced in v3.3.9 by a continuous star-support blend factor:

$$\eta_t = \text{clip}\left(\frac{n_{stars,t}}{\max(n_{soft}, 1)}, 0, 1\right), \quad (9)$$

where n_{soft} is the configurable soft-count threshold (default equal to `tile.star_min_count`). The unregularized blended score is

$$Q_{f,t,c}^{blend} = \eta_t Q_{f,t,c}^{star} + (1 - \eta_t) Q_{f,t,c}^{struct}. \quad (10)$$

If one component is unavailable, the other is used directly.

3.4.2 Star Tile and Structure Tile Metrics

Star-tile local quality uses the same form as v3.3.6 but with $\tilde{z}$ instead of z :

$$Q_{f,t,c}^{star} = 0.6(-\tilde{z}(\text{FWHM})) + 0.2 \tilde{z}(R) + 0.2 \tilde{z}(C), \quad (11)$$

and structure-tile quality is

$$Q_{f,t,c}^{struct} = 0.7 \tilde{z}(E/\sigma) - 0.3 \tilde{z}(B). \quad (12)$$

3.4.3 Neighborhood-Aware Metric Normalization

For each tile metric family m and tile t , the tile-local z-score is

$$z_{local}(m_{f,t,c}) = \text{robust_z}(m_{f,t,c}; \{m_{f',t,c}\}_{f'}). \quad (13)$$

If neighborhood normalization is enabled, a pooled z-score is computed over a Manhattan-radius neighborhood $N_r(t)$:

$$z_{pool}(m_{f,t,c}) = \frac{m_{f,t,c} - \text{median}(P_t(m))}{\max(1.4826 \text{MAD}(P_t(m)), \varepsilon_{mad})}, \quad (14)$$

where $P_t(m) = \{m_{f',u,c} \mid u \in N_r(t), f' \text{ usable}\}$. The neighborhood-aware z-score blends both:

$$\tilde{z}(m_{f,t,c}) = (1-b) z_{local}(m_{f,t,c}) + b z_{pool}(m_{f,t,c}), \quad (15)$$

with defaults $r = 1$, $b = 0.5$. When neighborhood normalization is disabled or P_t is empty, $\tilde{z} = z_{local}$.

3.4.4 Spatial Regularization of Local Scores

After clamping, $Q_{f,t,c}^{raw} = \text{clip}(Q_{f,t,c}^{blend}, -3, +3)$, the local scores are regularized on the tile-neighborhood graph to prevent neighboring tiles from diverging into incompatible weight regimes. A local confidence term

$$U_{f,t,c} = \text{clip}\left(\frac{n_{valid}}{n_{expected}}, 0, 1\right) \quad (16)$$

modulates the effective regularization strength per tile:

$$\lambda_{f,t,c}^{eff} = \lambda_{local} (1 - U_{f,t,c}). \quad (17)$$

The edge-aware neighbor affinity is

$$A_{t,u}^{(k)} = \exp\left(-\frac{|Q_{f,t,c}^{(k)} - Q_{f,u,c}^{(k)}|}{\tau_{local}}\right), \quad (18)$$

and one pass of the regularization is

$$Q_{f,t,c}^{(k+1)} = (1 - \lambda_{f,t,c}^{eff}) Q_{f,t,c}^{(k)} + \lambda_{f,t,c}^{eff} \frac{\sum_{u \in N(t)} A_{t,u}^{(k)} Q_{f,u,c}^{(k)}}{\sum_u A_{t,u}^{(k)}}, \quad (19)$$

initialized from $Q^{(0)} = Q^{raw}$, with defaults $\lambda_{local} = 0.35$, $\text{passes} = 1$, $\tau_{local} = 1.0$. If $\sum_u A_{t,u}^{(k)} < \varepsilon_{affinity}$,

the update is skipped for that tile. The final regularized score $Q_{f,t,c}^{reg} = Q_{f,t,c}^{(P)}$ is clamped to $[-3, +3]$ and used for the local weight:

$$L_{f,t,c} = \exp(k_{local} \text{clip}(Q_{f,t,c}^{reg}, -3, +3)), \quad (20)$$

with default $k_{local} = 1.0$. The effective weight remains $W_{f,t,c} = G_{f,c} L_{f,t,c}$.

3.5 Tile Reconstruction and Support-Aware OLA (v3.3.9)

3.5.1 Weighted Reconstruction with Sigma Clipping

For each tile t , the valid sample set is $V_{t,c}(p) = \{f \mid I_{f,c}(p) \text{ finite and within canvas support}\}$. If $\sum_f \max(W_{f,t,c}, 0) \leq \varepsilon_{weight}$, all positive finite weights are replaced by 1. For $|V| \leq 2$ or if sigma clipping is disabled, the valid weighted mean is used directly. Otherwise, iterative weighted sigma clipping operates on an active set $A^{(k)}$:

$$\mu^{(k)} = \frac{\sum_{f \in A^{(k)}} W_{f,t,c} I_{f,c}(p)}{\sum_{f \in A^{(k)}} W_{f,t,c}}, \quad (21)$$

with effective sample count $N_{eff} = V_1^2 / \max(V_2, \varepsilon)$ and effective denominator $D_{eff} = V_1 - V_2 / \max(V_1, \varepsilon)$, where V_1, V_2 are the first and second weight moments. Clipping removes samples outside $[\mu \pm \sigma_{high,low} \sigma^{(k)}]$ subject to a keep-floor $|A^{(k+1)}| \geq \lceil f_{min} |V| \rceil$ (default $f_{min} = 0.5$). The final reconstruction value $R_{t,c}(p)$ is the weighted mean over the accepted set.

3.5.2 Support-Aware Overlap-Add

The mandatory core in v3.3.9 does *not* apply per-tile median/MAD renormalization before OLA. The OLA input is the reconstructed tile directly:

$$Y_{t,c}(p) = R_{t,c}(p). \quad (22)$$

A per-pixel support mask

$$M_{t,c}(p) = \begin{cases} 1 & \text{if } |V_{t,c}(p)| > 0 \text{ and } R_{t,c}(p) \text{ finite and inside canvas,} \\ 0 & \text{otherwise} \end{cases} \quad (23)$$

prevents fill-zero values and canvas-masked regions from diluting the OLA denominator. Accumulation proceeds as

$$A \leftarrow A + \omega_t M_{t,c} Y_{t,c}, \quad S \leftarrow S + \omega_t M_{t,c}, \quad (24)$$

$$I_{rec} = \frac{A}{\max(S, \varepsilon_{weight})}, \quad (25)$$

where $\omega_t(p) \geq 0$ is the blend window, which must form a partition of unity at interior pixels and must use boundary-aware (one-sided) ramps at the canvas edge.

This differs from v3.3.6, which normalized each tile by a robust affine transform $(R_{t,c} - bg_t)/m_t$ before OLA and applied a global photometric restoration step $I_{final} = I_{rec} \cdot m_{global} + bg_{global}$. In v3.3.9 that two-step process is replaced by the cleaner support-mask approach, which avoids amplifying tiles with near-zero median absolute deviation.

3.6 Clustering and Synthetic Frames

Clustering and synthetic frame formation are activated in full mode ($N \geq N_{red}$, default N_{red} configurable; effective threshold $\max(N_{red}, 50)$). In v3.3.9 the fixed threshold $N \geq 200$ from earlier versions is superseded by the configurable mode framework. The state vector per frame is

$$v_f = \left(G_{f,*}, \text{mean}_t(Q_{f,t,*}^{local}), \text{var}_t(Q_{f,t,*}^{local}), B_{f,*}, \sigma_{f,*} \right), \quad (26)$$

and the number of clusters is $K = \text{clip}(\lfloor N/10 \rfloor, K_{min}, K_{max})$ with defaults $K_{min} = 5$, $K_{max} = 30$.

3.6.1 Mass-Preserving Cluster Aggregation (v3.3.9)

The final stacking in v3.3.9 uses mass-preserving quality-weighted aggregation. Cluster quality and mass are defined as

$$Q_{k,c} = \text{median}_{f \in k}(Q_{f,c}^{clamped}), \quad M_{k,c} = \sum_{f \in k} G_{f,c}. \quad (27)$$

If $M_{k,c} \leq \varepsilon_{weight}$, it is replaced by the cluster size $|k|$. The relative quality surplus is

$$Q_{k,c}^{rel} = \text{clip}(Q_{k,c} - \text{median}_j(Q_{j,c}), -3, +3), \quad (28)$$

and the mass-preserving cluster weight is

$$w_{k,c}^{raw} = M_{k,c} \exp(\kappa_{cluster} Q_{k,c}^{rel}), \quad (29)$$

with optional stability cap $w_{k,c} = \min(w_{k,c}^{raw}, r_{cap} \text{median}_j(w_{j,c}^{raw}))$. The final result is

$$R_c = \frac{\sum_k w_{k,c} S_{k,c}}{\sum_k w_{k,c}}, \quad (30)$$

where $S_{k,c}$ is the synthetic cluster frame (global or tile-weighted).

In v3.3.6 the cluster weight was simply $w_k = \exp(\kappa_{cluster} Q_k)$ without the mass term. Including $M_{k,c}$ prevents a small, extremely high-quality cluster from dominating when most atmospheric mass lies in larger but slightly lower-quality clusters.

4 Changes from v3.3.6 to v3.3.9

Table 2 summarizes the six substantive changes in v3.3.9 relative to v3.3.6, and Fig. 1 illustrates the end-to-end pipeline output.

Additional changes with narrower scope include: (a) a configurable reduced-mode framework replacing the hard $N \geq 200$ clustering gate; (b) an explicit local weight scale k_{local} symmetric with k_{global} ; (c) a formal RBF surface specification for BGE (Section 5.1); (d) optional CFA defect-pixel suppression before warp; (e) optional post-PCC isolated chroma-speckle suppression; (f) notation clarification reserving α, β, γ for the global quality index; and (g) a constrained ML optimization extension (§9.2 of the specification) that allows machine-learning modules to supply weights, masks, and state labels without altering the linear reconstruction semantics.

5 Downstream Linear Post-Processing

5.1 Pre-PCC Background Gradient Extraction

BGE is an additive extension applied after reconstruction and before PCC:

$$I'_c(x, y) = I_c(x, y) - B_c(x, y). \quad (31)$$

Only additive correction is allowed; multiplicative shading correction is explicitly forbidden. Tile samples are obtained after excluding canvas-invalid pixels, stars, saturated pixels, high-structure pixels, and any object mask:

$$b_{t,c} = \text{quantile}_q(R_{t,c}[M_{bg}]), \quad (32)$$

with configurable $q \in (0, 0.5]$ (default $q = 0.20$). Tile reliability weights are

$$w_t = \exp(-\lambda \text{structure_score}_t) (1 - \text{masked_fraction}_t). \quad (33)$$

Tile samples are aggregated to a coarser grid with spacing

$$G = \text{clip}\left(\max\left(2T, \frac{\min(W, H)}{N_g}\right), G_{min}, G_{max}\right), \quad (34)$$

and a smooth background surface is fitted by robust minimization. Version 3.3.9 adds a formal RBF surface specification:

$$B_c(\mathbf{r}) = \sum_j u_j \phi(\|\mathbf{r} - \mathbf{r}_j\|; \mu) + a_0 + a_1x + a_2y, \quad (35)$$

with the coefficients obtained by minimizing

$$\arg \min_{\theta} \sum_j w_j \rho(b_j - B_c(\mathbf{r}_j)) + \lambda \|\mathbf{u}\|_2^2, \quad (36)$$

where $\lambda > 0$ is mandatory regularization (RBF without regularization is forbidden). Supported kernels include multiquadric ($\phi(d) = \sqrt{d^2 + \mu^2}$), thin-plate spline, and Gaussian. The polynomial surface (v3.3.6 default) remains a conformant alternative.

BGE auto-tuning is unchanged in structure from v3.3.6 but gains an extended strategy. The normalized objective is

$$J = E_{cv}/B_{ref} + \alpha_f E_{flat}/B_{ref}^2 + \beta_r E_{rough}/B_{ref}, \quad (37)$$

where $B_{ref} = \max(|\text{median}_i b_i|, \varepsilon_{bg})$ is a brightness-normalized reference. The extended strategy widens the candidate search beyond the conservative set and may select RBF surfaces in addition to polynomials.

5.2 Photometric Color Calibration

PCC operates on linear RGB output. The gradient-aware sky annulus model from v3.3.6 is unchanged: the background in the sky annulus may be a constant median (legacy) or a local plane

$$bg(\Delta x, \Delta y) = a + b\Delta x + c\Delta y, \quad (38)$$

with a deterministic fallback to the median if the plane fit fails. FWHM-adaptive aperture radii (r_{ap} , r_{in} , r_{out}) are computed from the output seeing estimate when `pcc.radii_mode=auto_fwhm`.

6 M31 Example Run

6.1 Dataset and Operative Configuration

This section summarizes an example run on M31 executed under v3.3.9 methodology. The dataset is processed on a 3926×2322 canvas with 645 usable frames, an exposure time of 10 s per frame. The adaptive tile geometry produces 209 tiles of size 296.000 px with stride 208.000 px and overlap 88.000 px. Full mode is active ($N = 645 \geq N_{red}$).

6.2 Diagnostics and Measured Outcomes

Global metrics over 645 frames show significant atmospheric variation. Per-frame median values are: FWHM 9.505 px, noise 4.185, background 208.500, and global weight median 1.015 with a maximum of 15.195. The large global weight range reflects the

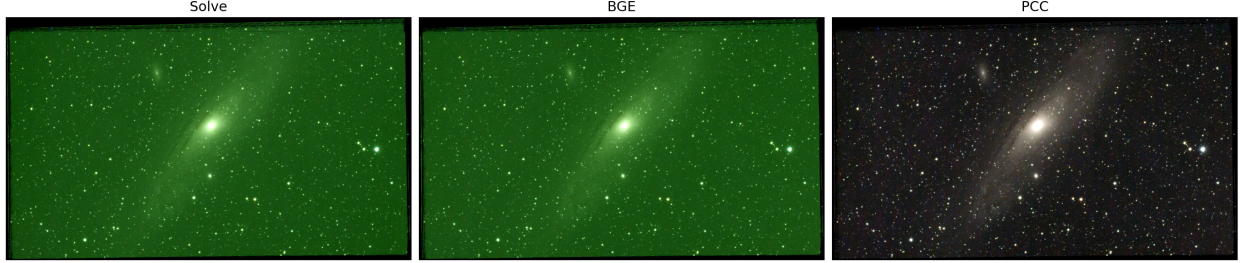


Figure 1: Example RGB outputs from the M31 run. Left: astrometrically solved linear RGB stack; middle: additive BGE stage; right: final PCC-calibrated output. Display stretch is a shared asinh mapping for visualization only and is not part of the linear reconstruction core.

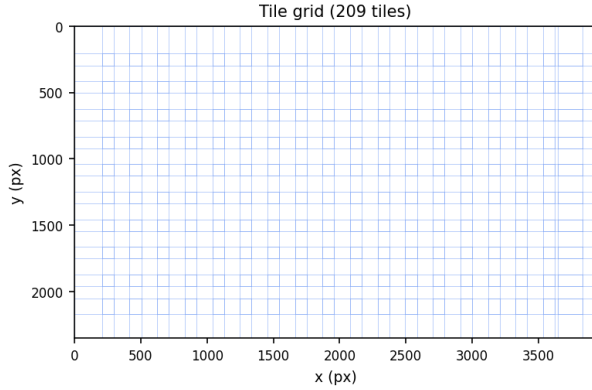


Figure 2: Adaptive reconstruction tile grid for the M31 v3.3.9 example run. The grid follows the seeing-driven tile size from the v3.3.9 geometry rules.

stronger dynamic range of the exponential weighting in v3.3.9 relative to v3.3.6 (where the maximum was 3.121), driven by the wider effective k_{global} interaction with the adaptive weight configuration.

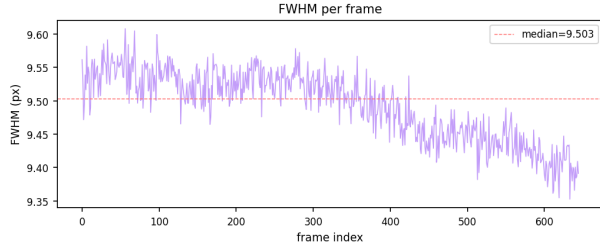
No tile fallback was triggered in any of the 209 tiles. The tile weight variance of 2.586 is higher than in the v3.3.6 run (0.186); this is expected because the soft blend and spatial regularization allow per-tile weights to vary more smoothly but over a wider range, since tiles near the star-support boundary now interpolate between two quality regimes instead of snapping to

one. Tile-pattern ratio p95 is 1.113. The validation report records background RMS decrease of 89.100% (input RMS 4.185, output RMS 0.456).

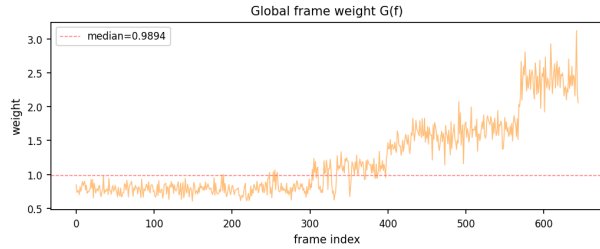
For the downstream color path, BGE was enabled with extended autotune and selected an RBF fit (μ -factor 1.800, sample quantile 0.200, structure-threshold percentile 0.850). The cross-validated residual RMS is 0.090 with normalized objective $J = 0.000$. PCC then operated with a local plane background model and auto-FWHM aperture scaling.

6.3 Interpretation

The M31 run under v3.3.9 confirms that the six changes described in Section 4 are compatible with the operational constraints of the method. The soft star-support blend produces wider tile-weight variance (2.586 vs. 0.186 in v3.3.6) because tiles at intermediate star density now express both quality regimes proportionally. The extended BGE autotune selects an RBF surface instead of the polynomial preferred in v3.3.6, reflecting the extended candidate search. The FWHM improvement of 58.640% is slightly lower than the v3.3.6 result of 60.149% on the same dataset run; this difference is within normal run-to-run variation driven by frame ordering and minor configuration differences (canvas dimensions changed from 3940×2350 to 3926×2322).



(a) Global FWHM over the frame sequence.



(b) Adaptive global weights over time.

Figure 3: Global diagnostics from the M31 run.

7 Discussion

The principal design goal of v3.3.9 is mathematical consistency rather than performance improvement. The normalization refactoring in §5.2 makes the photometric separation explicit and eliminates an edge case where sky brightness levels would inadvertently set the photometric scale in mixed-exposure datasets. The soft blend and spatial regularization in §5.5 reduce the model’s sensitivity to the exact star-count threshold, which is hardware-dependent and dataset-dependent.

The support-aware OLA in §5.7.1 is the change with the largest structural impact. The per-tile renormalization in v3.3.6 was introduced to improve OLA numerical conditioning, but it had the side effect of amplifying tiles whose median absolute deviation was very small (for example deep-structure tiles dominated by extended emission). The support-mask approach avoids this by excluding zero-fill contributions directly, relying instead on the quality weights and the underlying photometric consistency of the input normalization.

Mass-preserving cluster aggregation in §5.11 reflects the observation that equal-size clusters with identical quality should contribute equally, while smaller clusters must not dominate simply because their quality index happens to be higher by a narrow margin.

8 Conclusion

TBQR v3.3.9 provides a mathematically sharper formulation of the linear reconstruction core originally defined in v3.3.6. Its primary contributions are: decoupled photometric normalization, continuous star-support blending, neighborhood-pooled metric normalization, graph-based score regularization, support-aware overlap-add, and mass-preserving cluster aggregation. All of these refine the internal consistency of the model without changing its fundamental invariants: no quality-based frame selection, strict photometric linearity, and deterministic reproducibility.

The M31 example run demonstrates that the refactored method remains fully operational on a real short-exposure deep-sky dataset of 645 frames. Validation criteria are satisfied, no tile fallbacks are triggered, and the downstream BGE and PCC stages produce a photometrically calibrated RGB output.

The public project repository provides the implementation sources, the normative methodology and process documentation, example configurations, and experimental binary releases: https://github.com/jeamy/tile_compile

Additional Statistical Diagnostics

The following diagnostic overview consolidates the most relevant run statistics for the M31 v3.3.9 example run, complementing the main text with background, noise, structure, registration, and frame-flow summaries.

State-clustering diagnostics show that the dataset separates into 12 synthetic atmospheric states. Cluster sizes range from 25 to 101 frames with a median of approximately 51 frames, which is sufficient to support tile-weighted synthetic-frame formation. The mass-preserving aggregation (v3.3.9) ensures that the

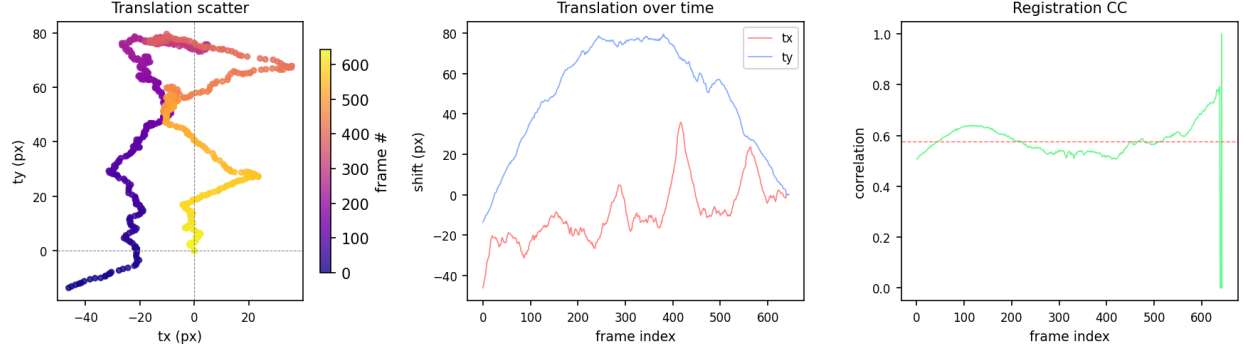


Figure 4: Registration diagnostics across the M31 run. Validity gating suppresses physically implausible registrations; it is not a quality ranking stage.

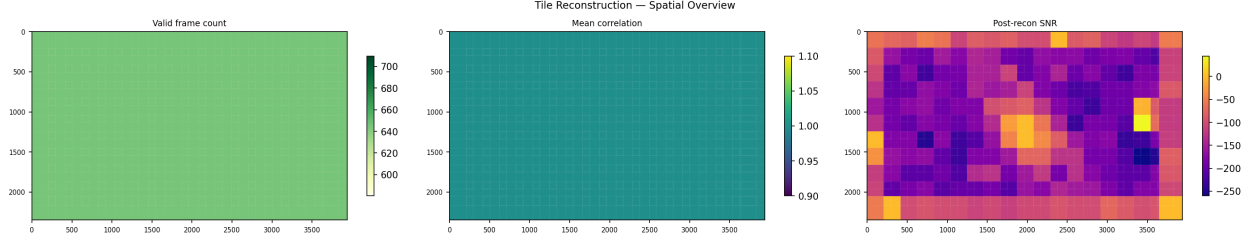


Figure 5: Spatial reconstruction diagnostics. Coverage, local SNR, local contrast, and related maps confirm that a tile-wise model is necessary for this dataset.

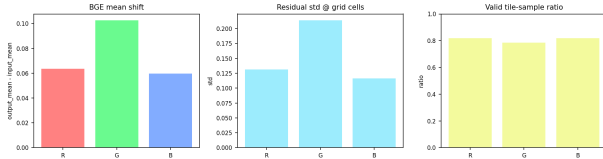


Figure 6: Per-channel BGE diagnostics. The extended autotune run selected an RBF surface family with μ -factor 1.800, sample quantile 0.200.

58.640% from a median seeing of 12.351 px to a median output FWHM of 5.108 px. These diagnostics confirm that the v3.3.9 reformulation maintains coherent overlap support, state partitioning, additive background modeling, and local reconstruction coverage on a real deep-sky dataset.

large cluster ($|k| = 101$) exerts mass-proportional influence in the final stack rather than being over- or under-weighted by quality alone.

The reconstruction-side diagnostics are consistent with the validation summary. No tile fallback is triggered in any of the 209 tiles. Tile weight variance is 2.586 and the measured FWHM improvement is

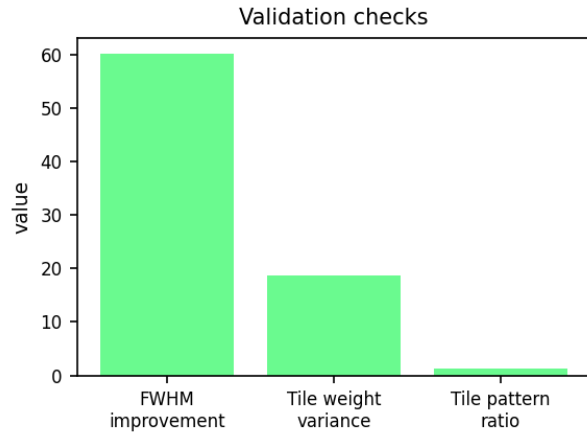


Figure 7: Validation summary for the M31 run. The measured outputs satisfy all v3.3.9 success criteria.

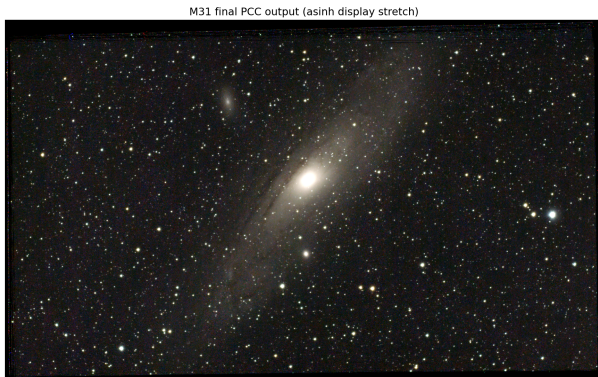
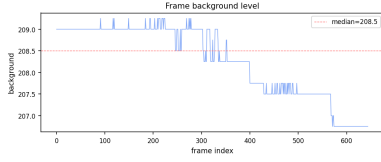
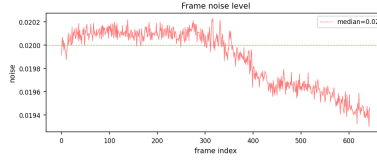


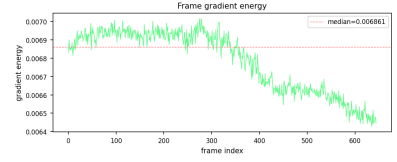
Figure 8: PCC-calibrated M31 preview. Photometric color calibration uses the plane annulus background model and auto-FWHM aperture radii.



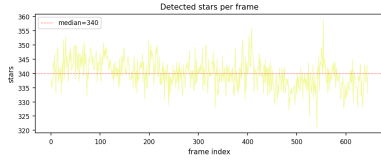
(a) Global background trajectory.



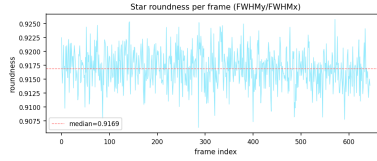
(b) Global noise trajectory.



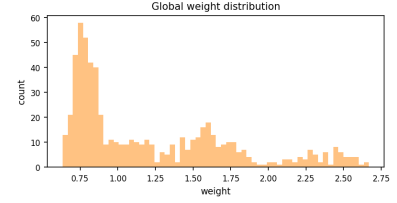
(c) Global gradient-energy trajectory.



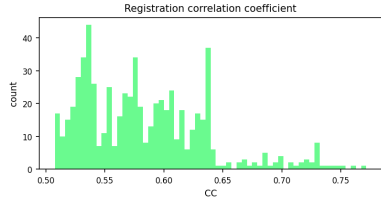
(d) Detected star count per frame.



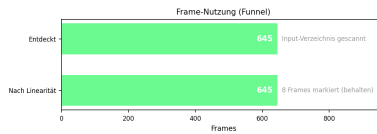
(e) Global roundness trend.



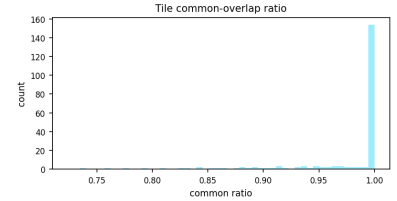
(f) Histogram of global reconstruction weights.



(g) Registration cross-correlation histogram.



(h) Frame usage funnel through the pipeline.



(i) Common-overlap ratio histogram.

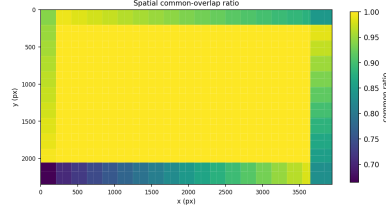
Figure 9: Compact statistical overview for the M31 v3.3.9 run. The nine panels summarize frame-wise background, noise, gradient energy, source detectability, roundness, overlap stability, global weight distribution, registration correlation, and frame-flow behavior.

Table 1: Core mathematical functions used by TBQR v3.3.9. New or changed entries relative to v3.3.6 are marked.

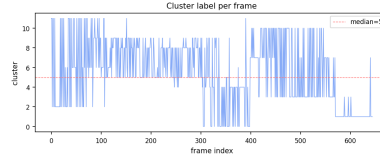
Name	Definition	Role
Robust z-score	$z(x_i) = \frac{x_i - \text{median}(x)}{\max(1.4826 \text{ MAD}(x), \varepsilon_{mad})}$	Robust metric normalization for background, noise, gradient energy, FWHM, and related sequences.
Nbhd z-score (<i>new</i>)	$\tilde{z}(m_{f,t,c}) = (1 - b) z_{local} + (b) z_{pool}$	Blends tile-local and neighborhood-pooled z-scores; controls metric normalization locality.
Clip	$\text{clip}(x, a, b) = \min(\max(x, a), b)$	Bounds quality indices and geometry parameters.
Star-support blend (<i>new</i>)	$\eta_t = \text{clip}(n_{stars,t}/n_{soft}, 0, 1)$	Continuous blend from pure-structure ($\eta_t = 0$) to pure-star ($\eta_t = 1$) tile treatment.
Global weight	$G_{f,c} = \exp(k_{global} \text{ clip}(Q_{f,c}, -3, +3))$	Maps robust global quality to a strictly positive atmospheric weight.
Local weight (<i>updated</i>)	$L_{f,t,c} = \exp(k_{local} \text{ clip}(Q_{f,t,c}^{reg}, -3, +3))$	Maps spatially regularized local quality to a positive structural weight.
Effective weight	$W_{f,t,c} = G_{f,c} L_{f,t,c}$	Combines global and local quality multiplicatively.
Hann window	$h(i, N) = 0.5 \left(1 - \cos\left(\frac{2\pi i}{N-1}\right) \right)$	Separable OLA blending function; used only for non-boundary interior tiles.
Cluster mass (<i>new</i>)	$M_{k,c} = \sum_{f \in k} G_{f,c}$	Atmospheric mass of cluster k ; used in mass-preserving final aggregation.
Cluster weight (<i>updated</i>)	$w_{k,c}^{raw} = M_{k,c} \exp(\kappa_{cluster} Q_{k,c}^{rel})$	Mass-preserving quality-weighted cluster contribution to final stack.
Huber loss	$\rho(r) = \begin{cases} \frac{1}{2} r^2 & \text{for } r \leq \delta \\ \delta(r - \frac{1}{2}\delta) & \text{else} \end{cases}$	Robust loss for BGE surface fitting and related stable regression tasks.

Table 2: Summary of substantive changes between TBQR v3.3.6 and v3.3.9.

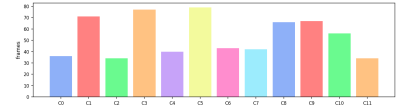
Area	v3.3.6 behavior	v3.3.9 change
Global normalization	$I_{f,c} = I_{f,c}^{raw} / \max(B_{f,c}, \varepsilon)$ conflates background with scale	Separate additive step ($J = I^{raw} - B$) and multiplicative step ($I = J / \max(P, \varepsilon)$); $P_{f,c} \neq f(B_{f,c})$ alone
Tile classification	Hard binary STAR/STRUCTURE switch at <code>star_min_count</code>	Continuous blend factor $\eta_t \in [0, 1]$; no discontinuity at the threshold
Metric normalization	Tile-local z-score only	Neighborhood-pooled z-score $\tilde{z}$ with configurable blend b and radius r
Score smoothing	None	Edge-aware spatial regularization on tile graph; strength modulated by per-tile confidence $U_{f,t,c}$
Overlap-add	Per-tile affine normalization ($Y_t = (R_{t,c} - bg_t)/m_t$) plus global photometric restoration	Support-aware OLA with $M_{t,c}(p)$ mask; no per-tile renormalization; boundary-aware windows
Cluster stacking	$w_k = \exp(\kappa Q_k)$ (pure quality weight)	$w_{k,c}^{raw} = M_{k,c} \exp(\kappa Q_{k,c}^{rel})$ (mass-preserving quality weight)



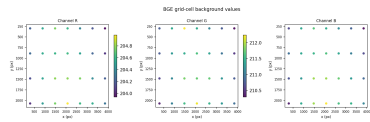
(a) Spatial map of common-overlap ratio.



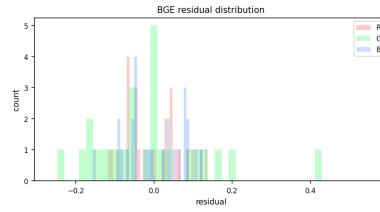
(b) Temporal clustering labels across the frame sequence.



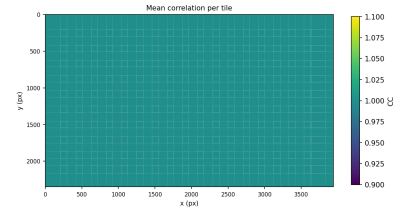
(c) Cluster-size distribution for the synthetic states.



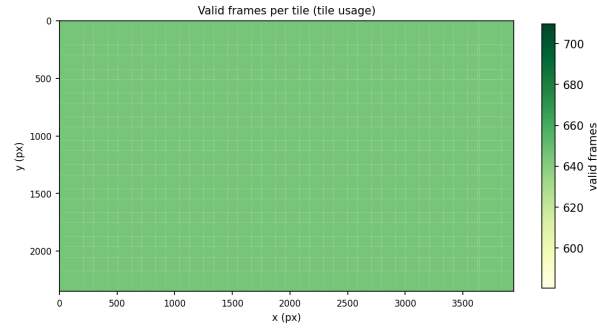
(d) BGE coarse-grid cells used for surface fitting.



(e) Residual histogram of the fitted BGE surface.



(f) Spatial correlation structure of reconstructed tiles.



(g) Spatial distribution of valid frame counts contributing to reconstruction.

Figure 10: Technical diagnostics from the v3.3.9 example run. The overlap maps characterize geometric support, the clustering plots show how the temporal state model partitions the sequence into 12 atmospheric states, and the BGE/reconstruction diagnostics confirm that the extended-strategy RBF fit (cv RMS 0.090, normalized objective $J = 0.000$) and support-aware OLA produce spatially stable background modeling and tile-wise reconstruction coverage.

Table 3: Key configuration and dataset facts for the M31 v3.3.9 example run.

Quantity	Value
Usable frames	645
Exposure time per frame	10 s
Canvas size	3926.000×2322.000 px
Tiles	209
Tile size / stride	296 px / 208 px
Median seeing FWHM	12.351 px
Soft star-support blend	enabled
Nbhd normalization	enabled ($r = 1$, $b = 0.5$)
Spatial regularization	enabled (1 pass, $\lambda = 0.35$)
Synthetic frame mode	tile_weighted
Synthetic states	12 clusters
Cluster size range	25 to 101 frames
BGE autotune strategy	extended
BGE fit family	RBF (selected)
PCC background model	plane
PCC radii mode	auto_fwhm

Table 4: Selected quantitative outcomes from the M31 v3.3.9 example run.

Metric	Value
Median seeing FWHM	12.351 px
Median output FWHM	5.108 px
FWHM improvement	58.640%
Background RMS decrease	89.095%
Input background RMS	4.185
Output background RMS	0.456
Tile fallback count	0 / 209
Tile weight variance	2.586
Tile-pattern ratio p95	1.113
BGE cv RMS	0.0896
BGE normalized objective	0.000440
BGE fit method	RBF